

	1	2	3	4	5	
A	12V Power Input File: POS12_power_input.kicad_sch 12V Telemetry File: POS12_telemetry.kicad_sch 3.3V Power Supply File: POS3P3_power_supply.kicad_sch 3.3V Telemetry File: POS3P3_telemetry.kicad_sch dsPIC33 Programming File: dspic33_programming.kicad_sch dsPIC33 Power File: dspic33_power.kicad_sch dsPIC33 GPIO File: dspic33_gpio.kicad_sch USB UART Bridge File: usb_uart_bridge.kicad_sch		Platform ETC File: platform_etc.kicad_sch PGOOD LEDs File: pgood_leds.kicad_sch Status LEDs File: status_leds.kicad_sch RCA Signal Conditioning File: rca_signal_conditioning.kicad_sch Speaker Signal Conditioning File: speaker_signal_conditoning.kicad_sch Source Selection, PGAs File: source_selection_pgas.kicad_sch Anti-alias Filtering File: anti_alias_filtering.kicad_sch Audio Voltage References File: audio_voltage_references.kicad_sch LED Panel Connectors File: led_panel_connectors.kicad_sch Pushbuttons File: pushbuttons.kicad_sch Mechanical File: mechanical.kicad_sch			
B						
C						
D						
	1	2	3	4	5	

Sheet: / File: Audio_Visualiser_DSP.kicad_sch		
Title:		
Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 1/20

TRY TO TALLY LOAD CURRENTS

TP1

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TP11

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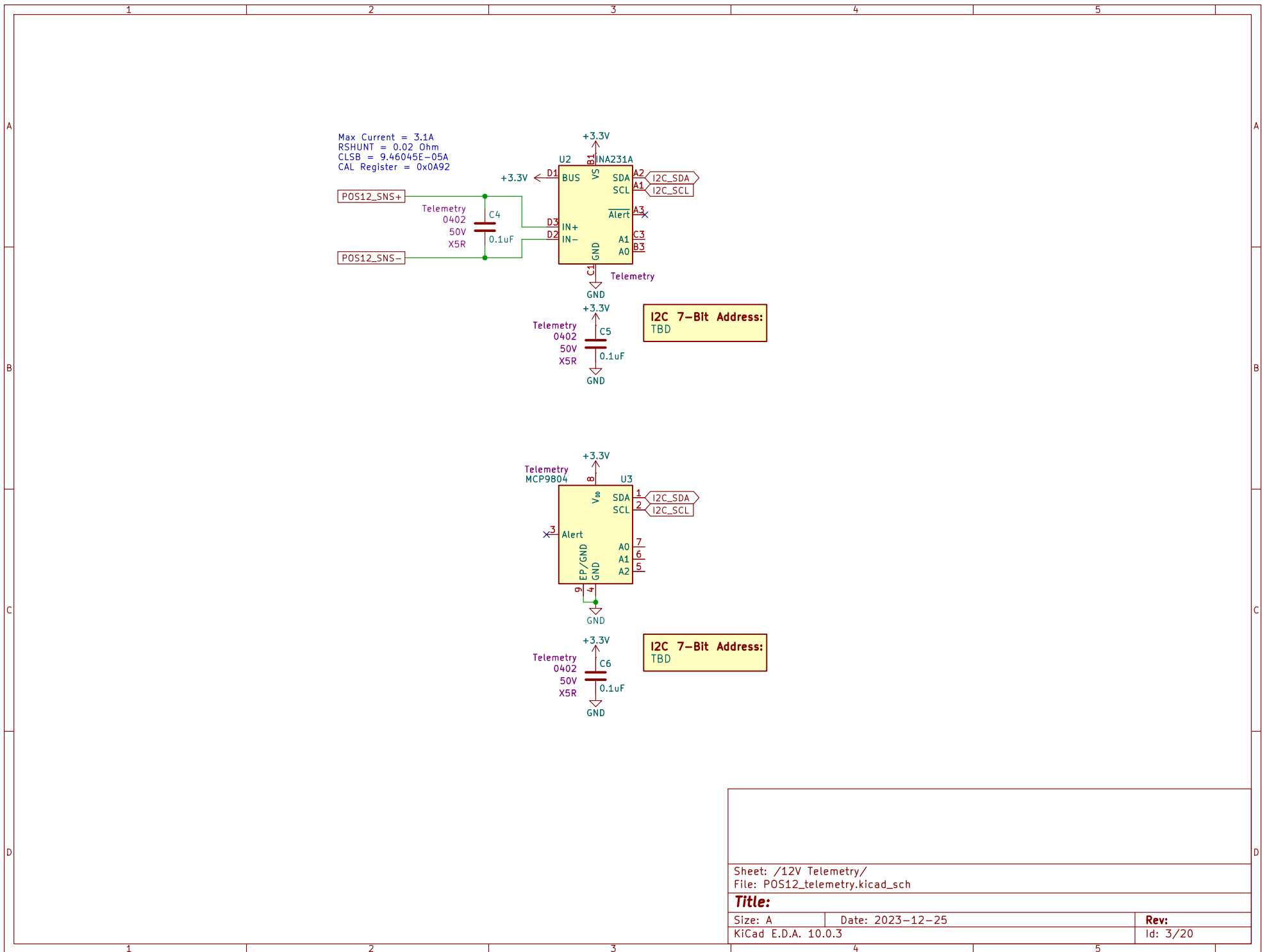
TP44

Title:		
Size: A	Date: 2023-12-25	Rev:
KICad E.D.A. 10.0.3		Id: 2/20

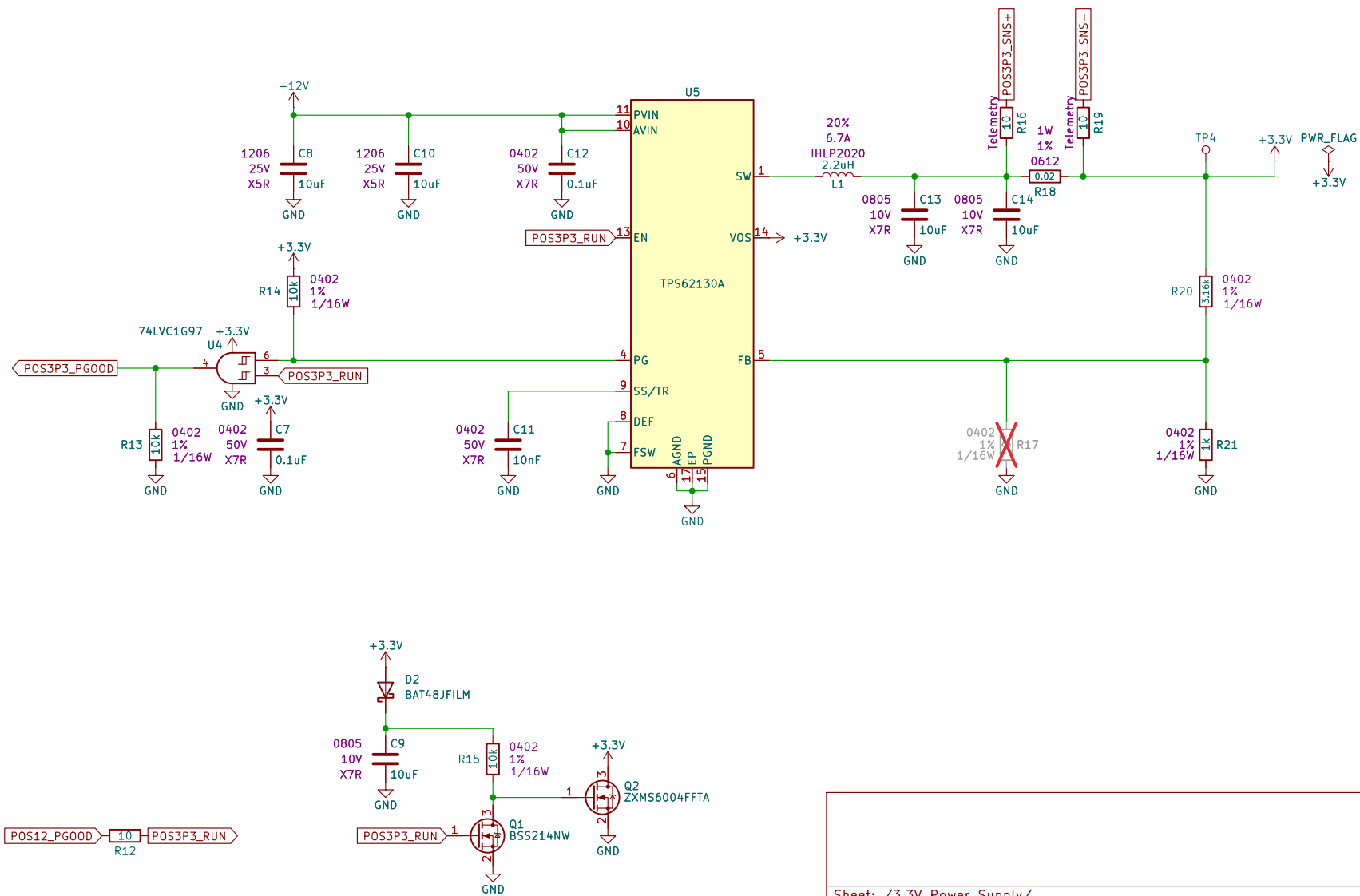
Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 2/20

KiCad E.D.A. 10.0.3	Id: 2/20
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[illegible][illegible]



TRY TO TALLY LOAD CURRENTS



Sheet: /3.3V Power Supply/
File: POS3P3_power_supply.kicad_sch

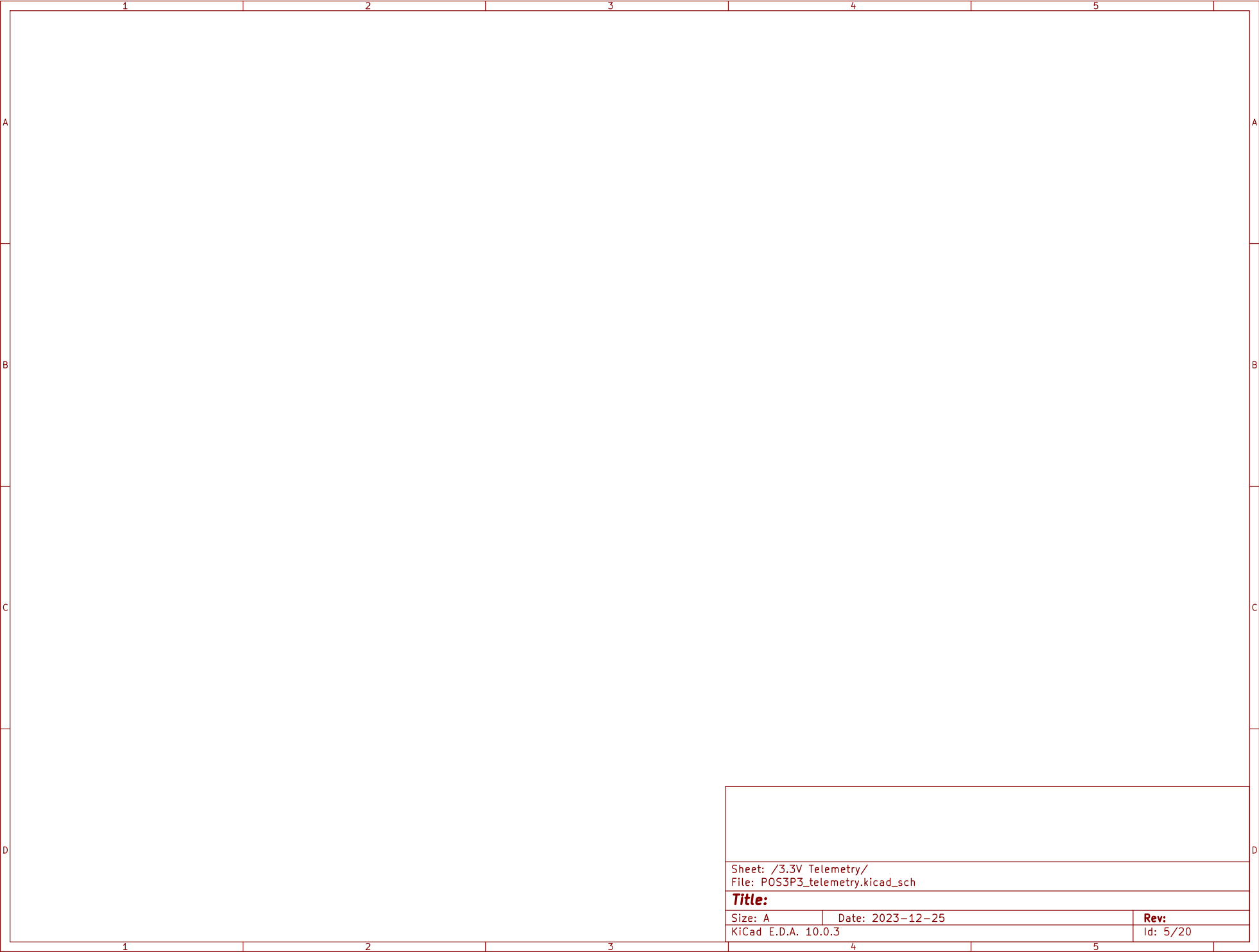
Title:

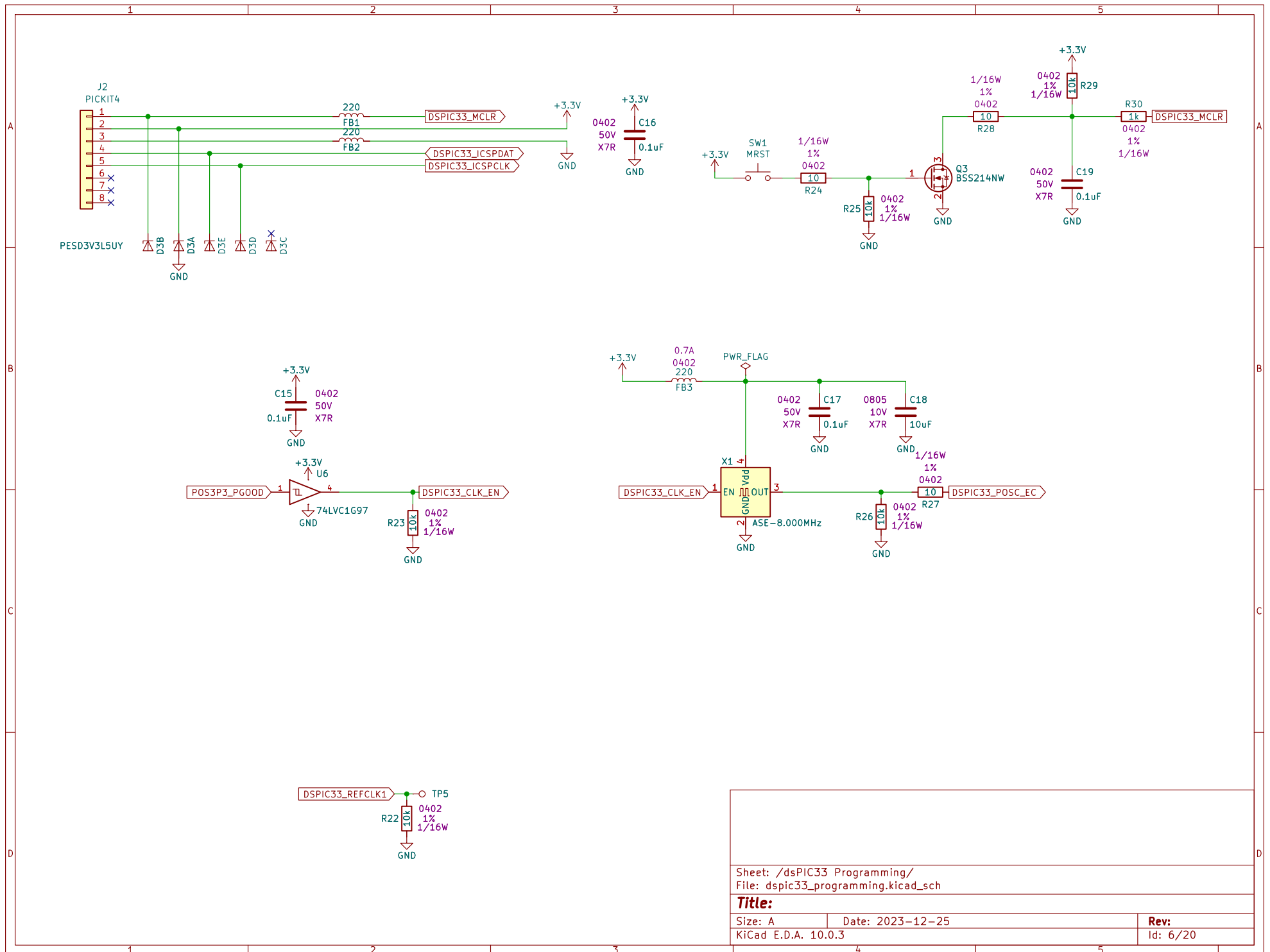
Size: A Date: 2023-12-25

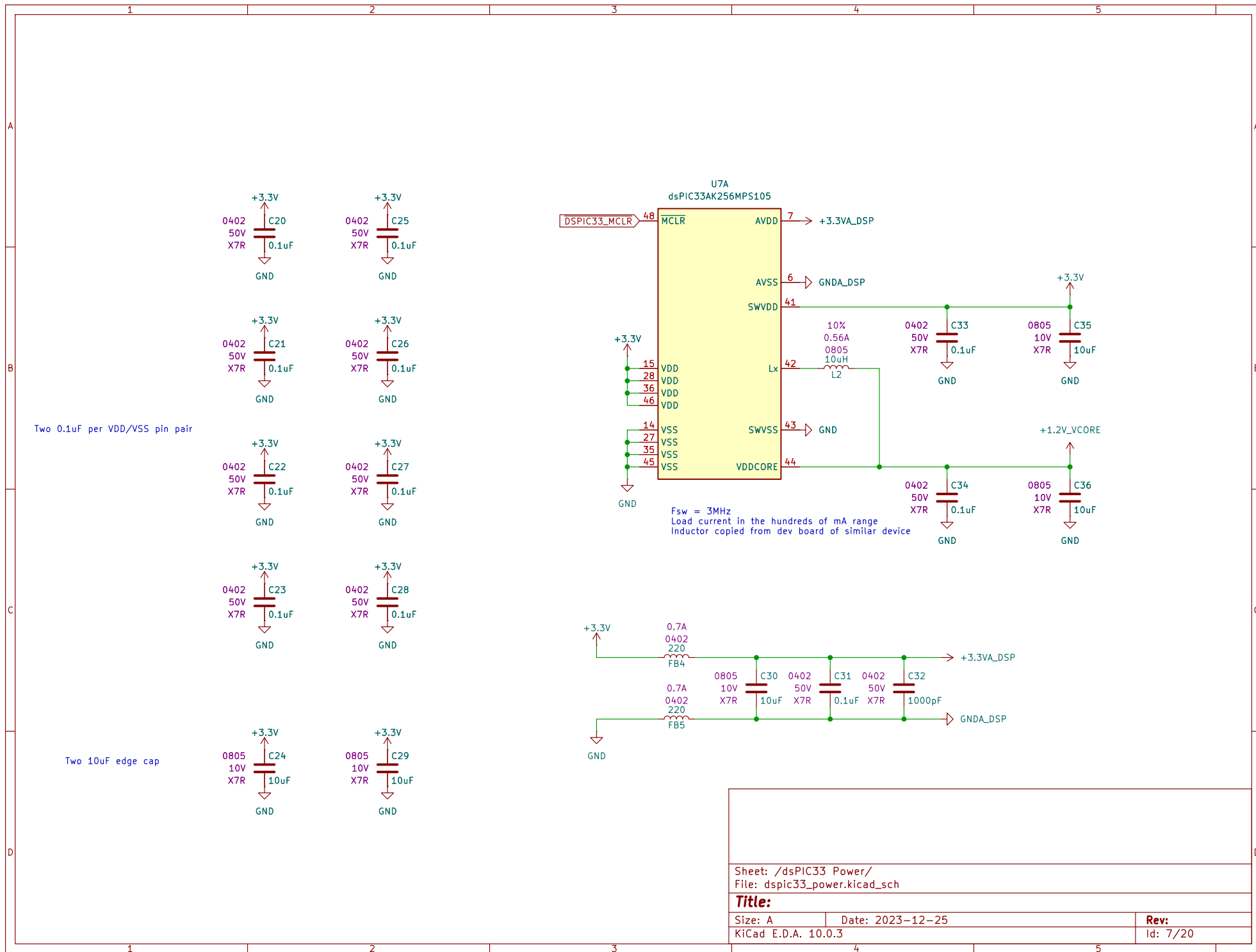
KiCad E.D.A. 10.0.3

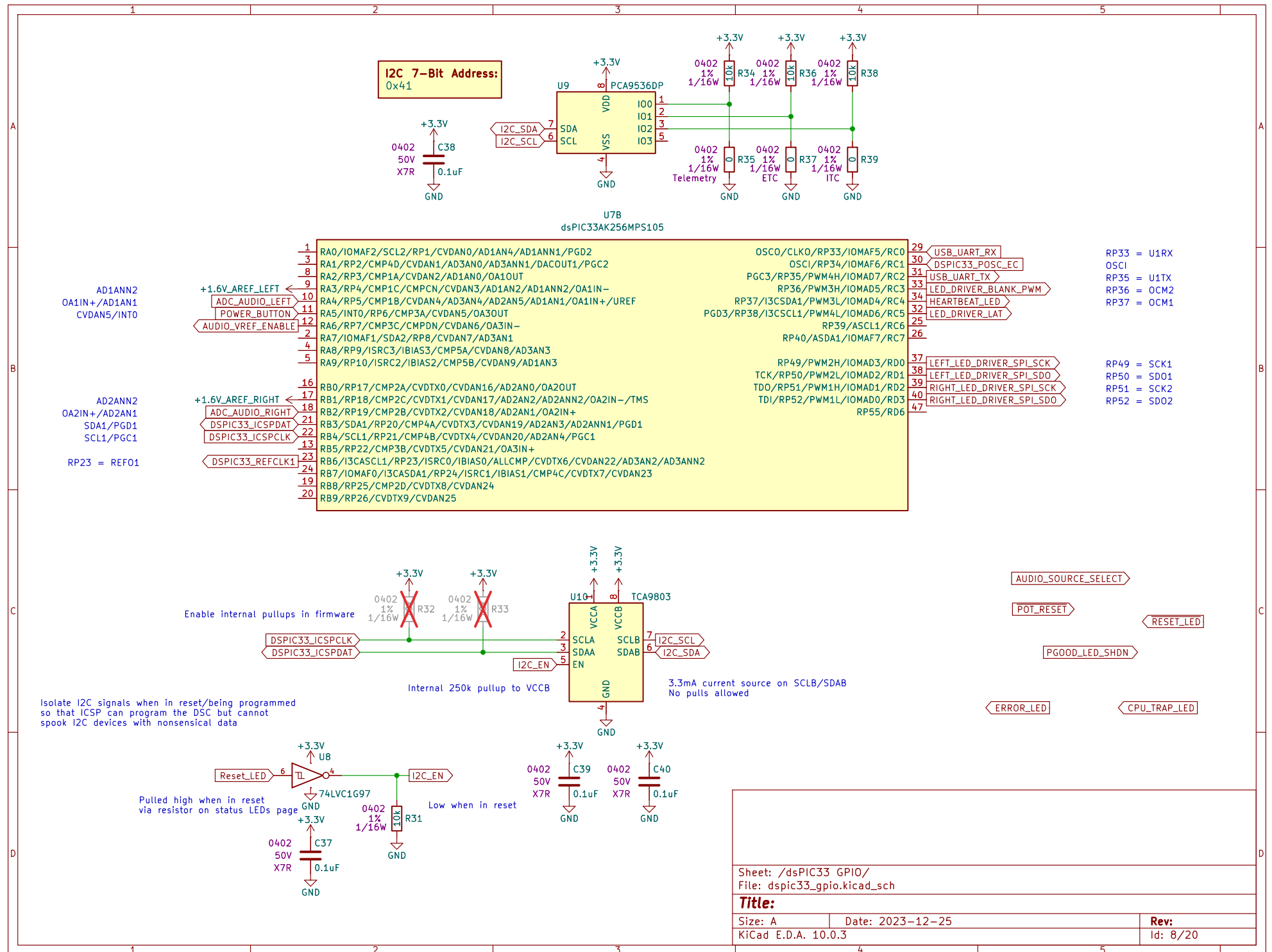
Rev:

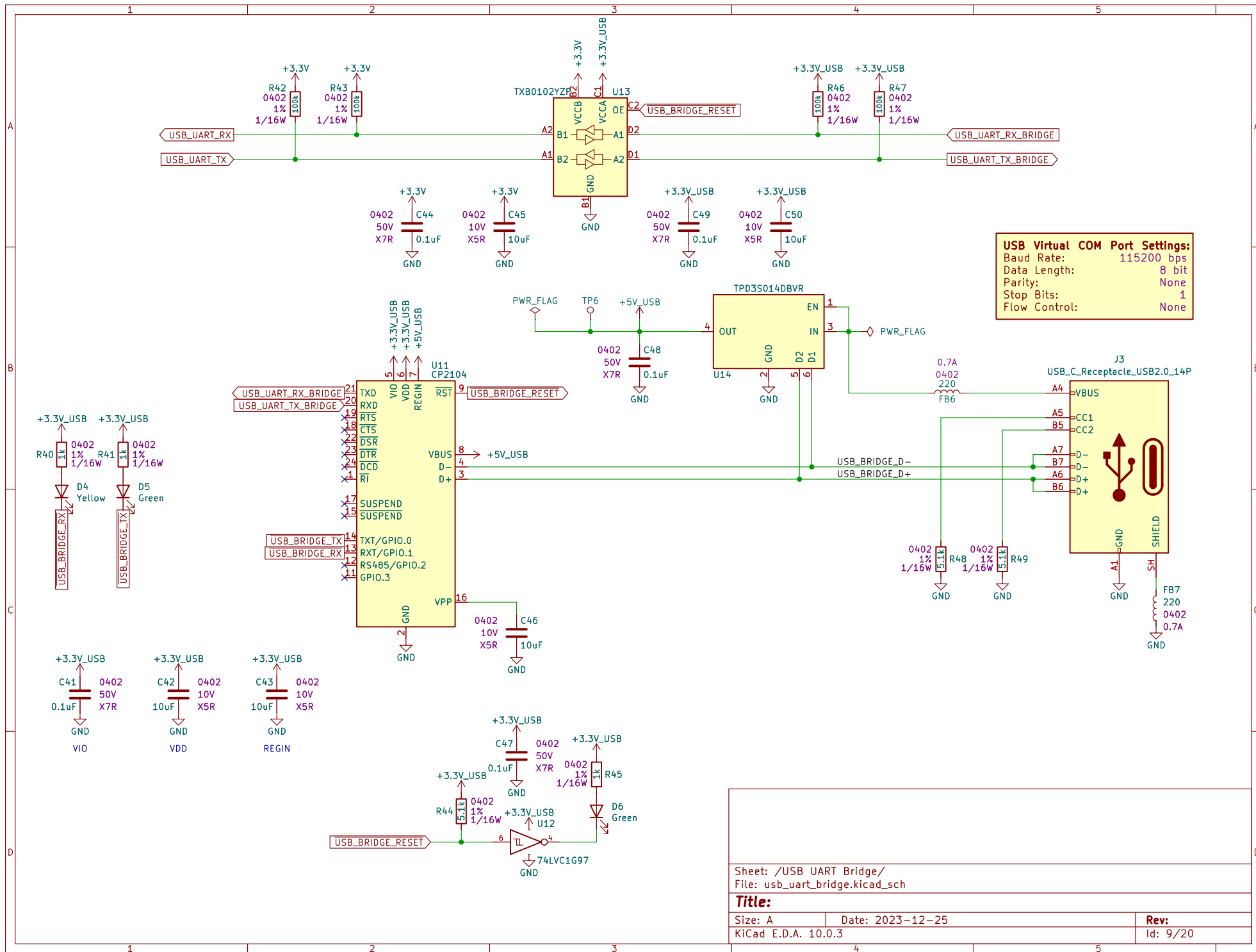
Id: 4/20









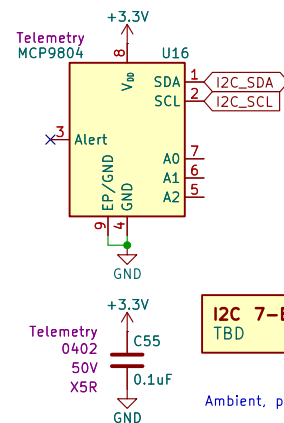
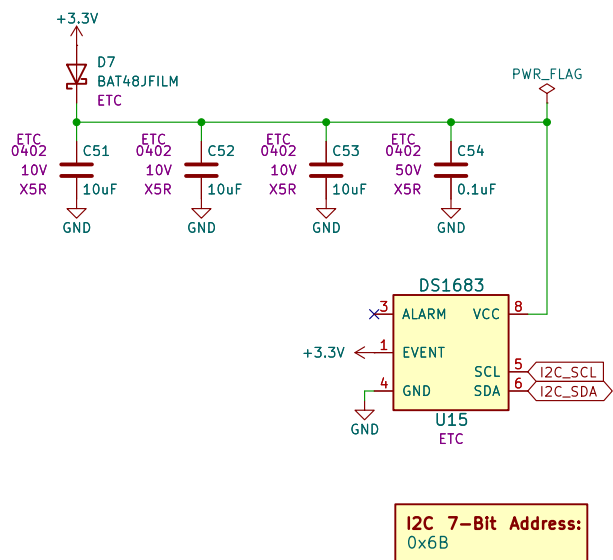


Sheet: /USB UART Bridge/
File: usb_uart_bridge.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 9/20



I2C 7-Bit Address:
TBD

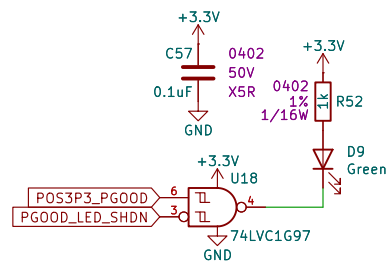
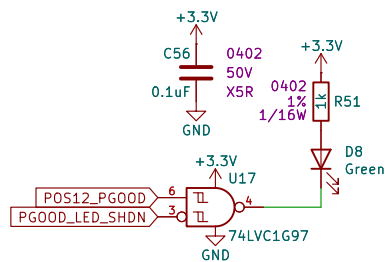
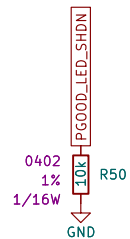
Ambient, place away from power components

Sheet: /Platform ETC/
File: platform_etc.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 10/20

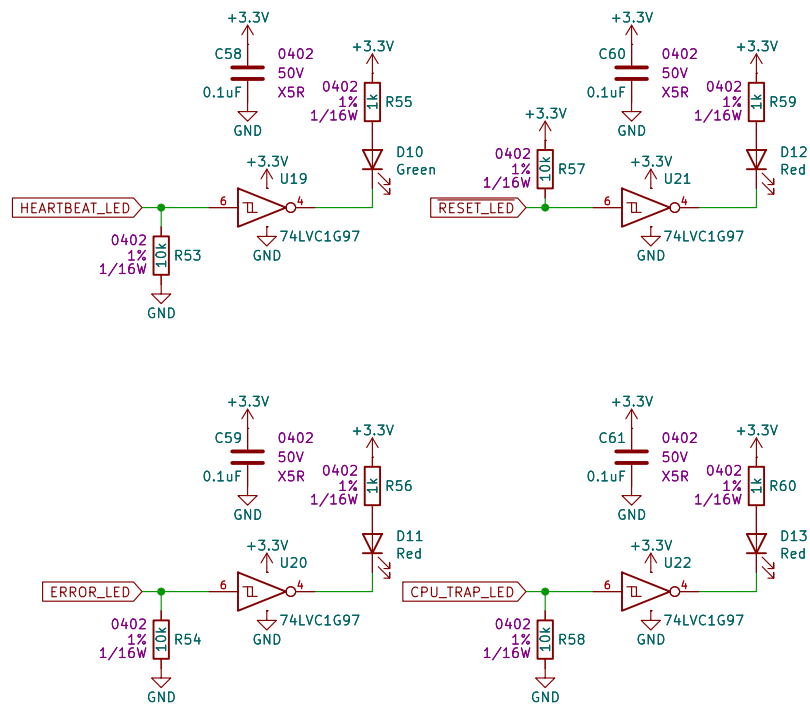


Sheet: /PGOOD LEDs/
File: pgood_leds.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 11/20

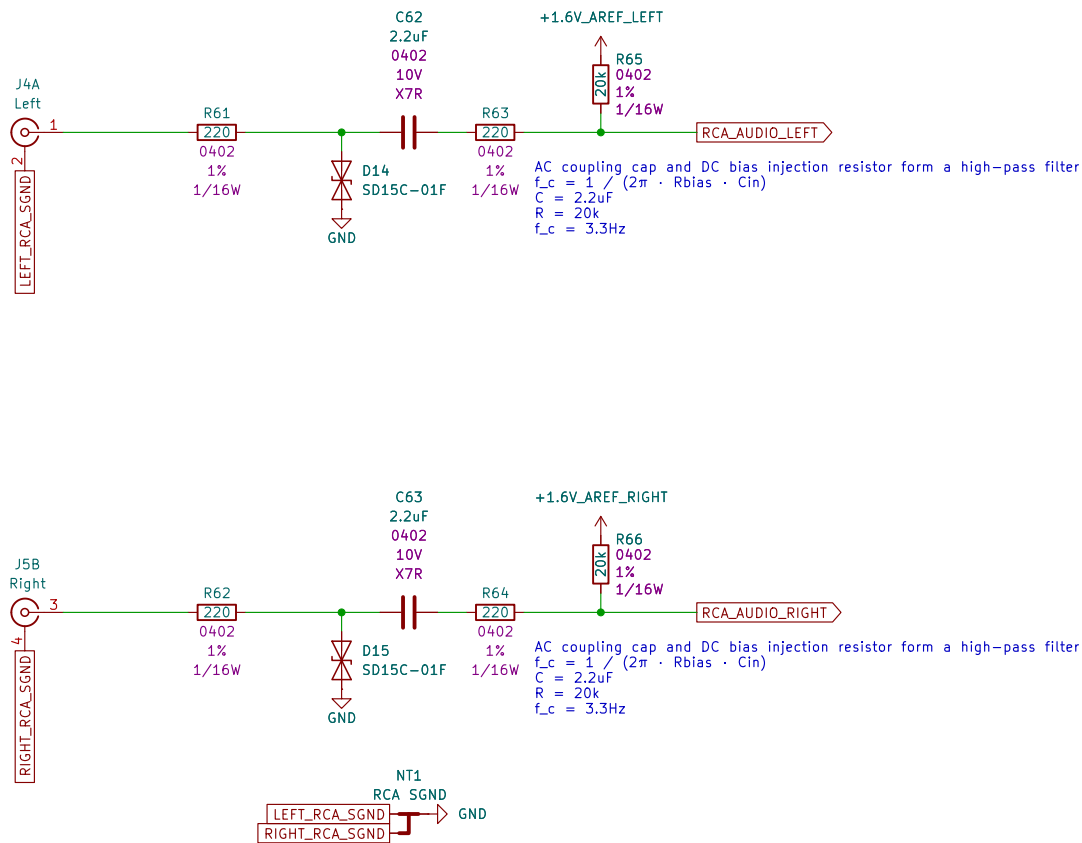


Sheet: /Status LEDs/
File: status_leds.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 12/20



Sheet: /RCA Signal Conditioning/
File: rca_signal_conditioning.kicad_sch

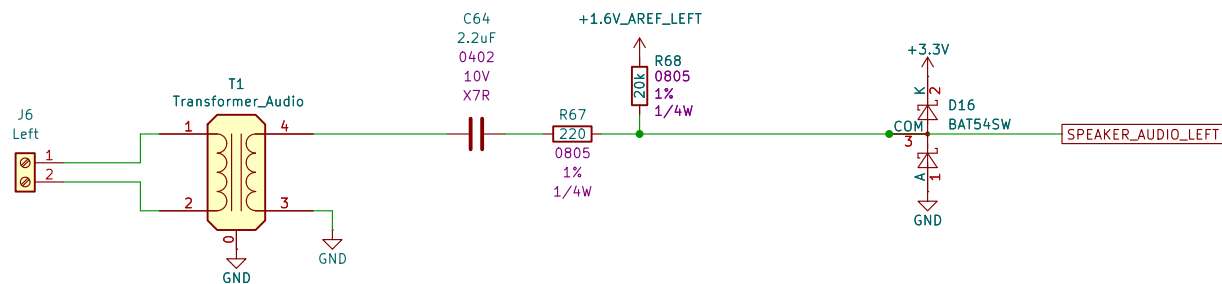
Title:

Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

Rev:
Id: 13/20

Alternate is a 20:1 resistive attenuation (for 18V peaks) and a diff amp



Sheet: /Speaker Signal Conditioning/
File: speaker_signal_conditioning.kicad_sch

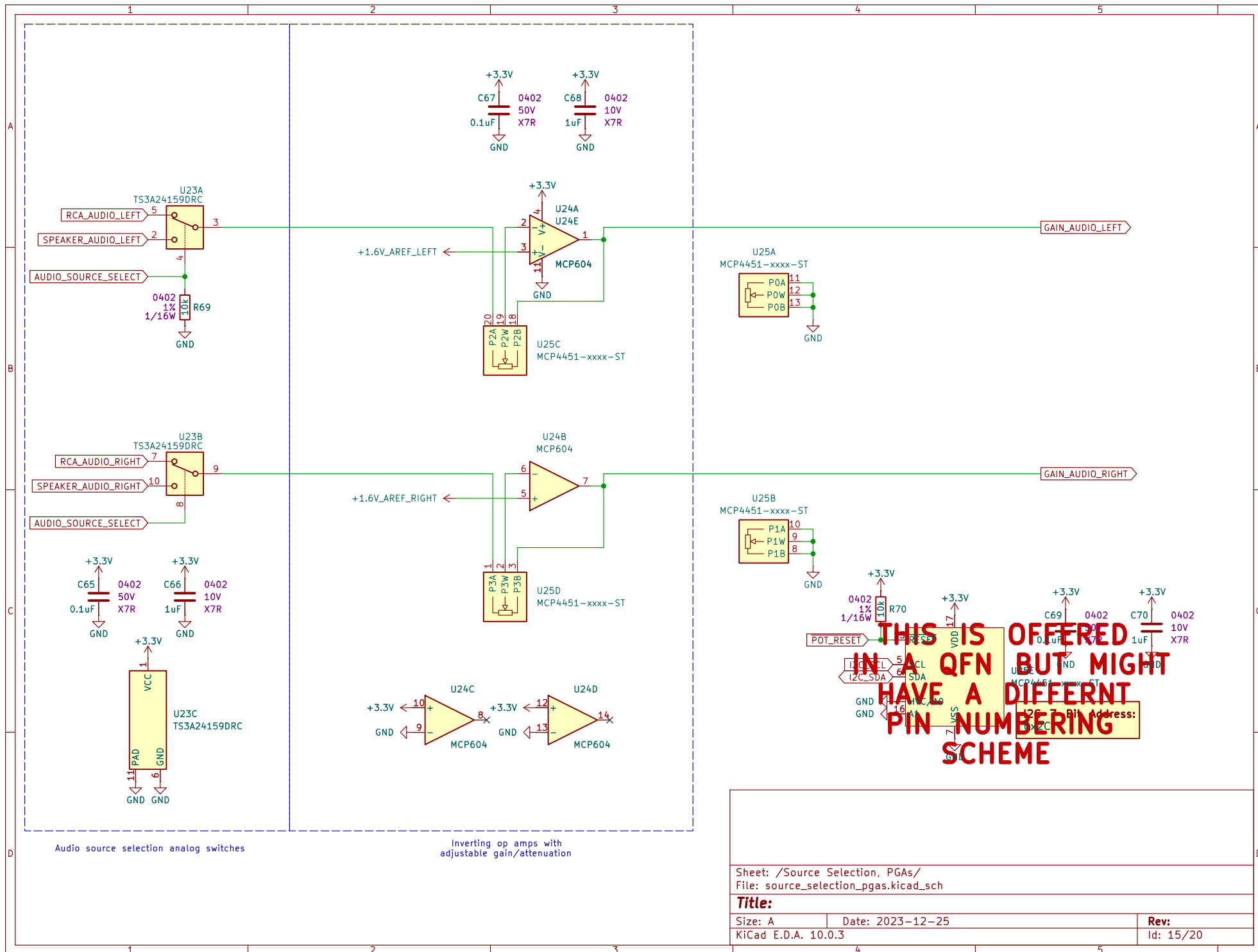
Title:

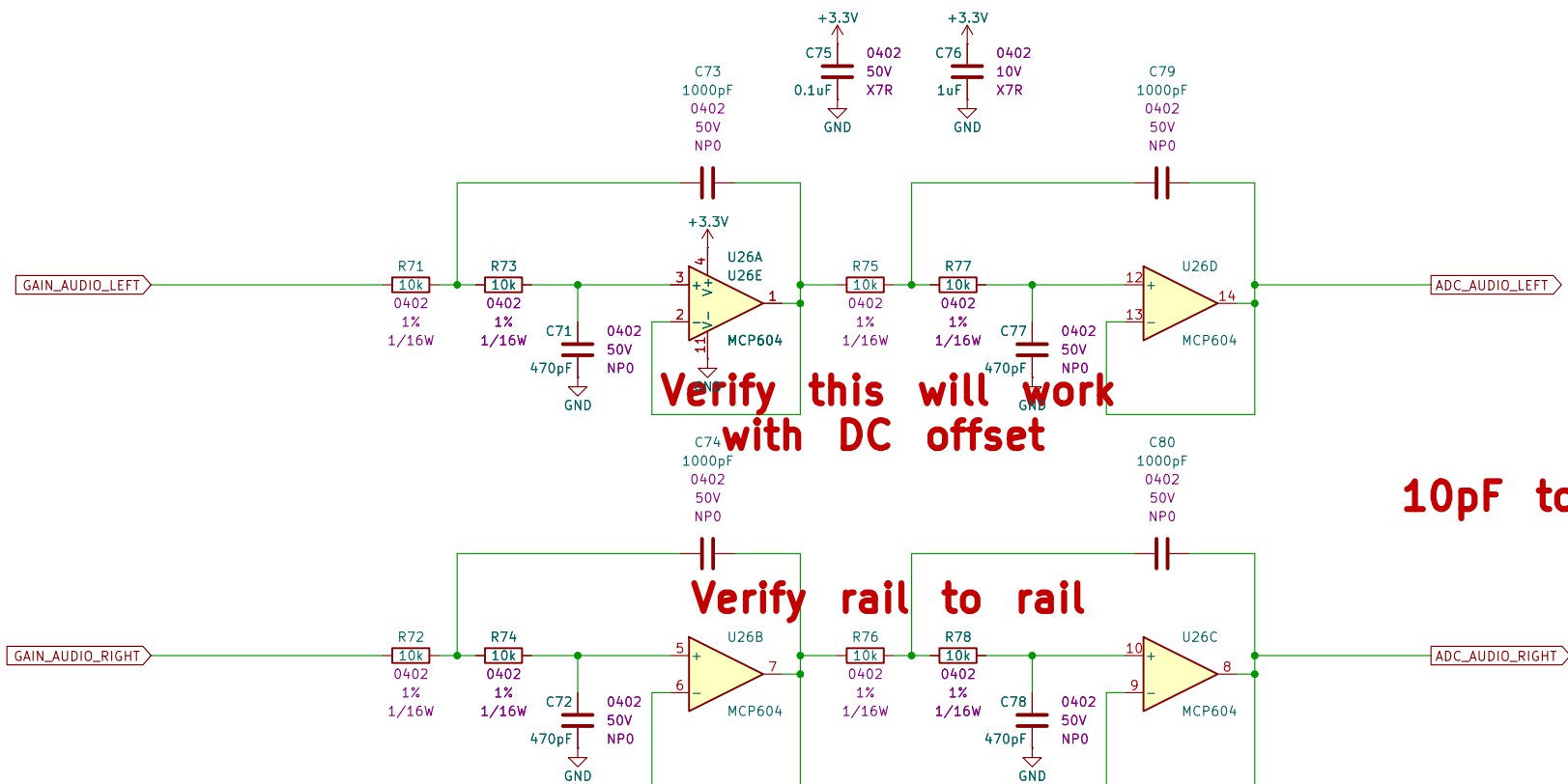
Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

Rev:

Id: 14/20





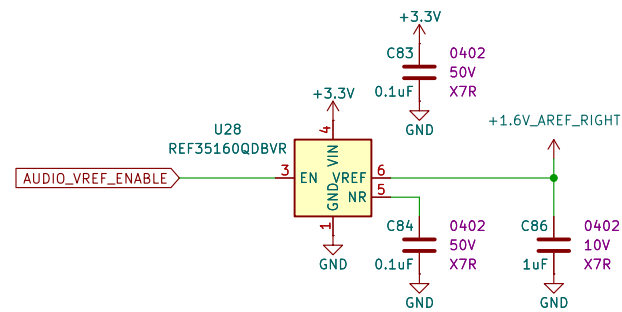
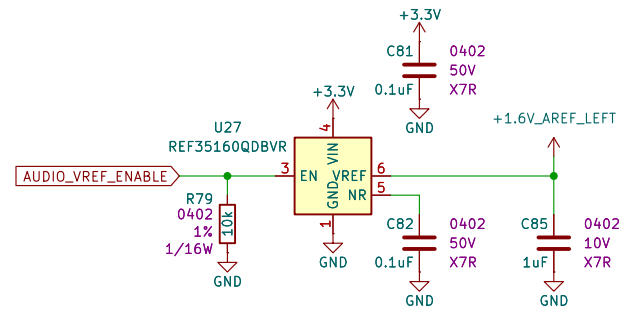
4-pole Sallen-Key active anti-aliasing lowpass filters
 Unity gain
 $f_c = 23.2\text{kHz}$, $Q = 0.707$ each
 If sampling at 44.1kHz , nyquist is 22.05kHz
 Must filter hard just below this

Sheet: /Anti-alias Filtering/
 File: anti_alias_filtering.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 16/20



Sheet: /Audio Voltage References/
File: audio_voltage_references.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 17/20

1					2					3					4					5					
A																									A
B																									B
C																									C
D																									D
1					2					3					4					5					

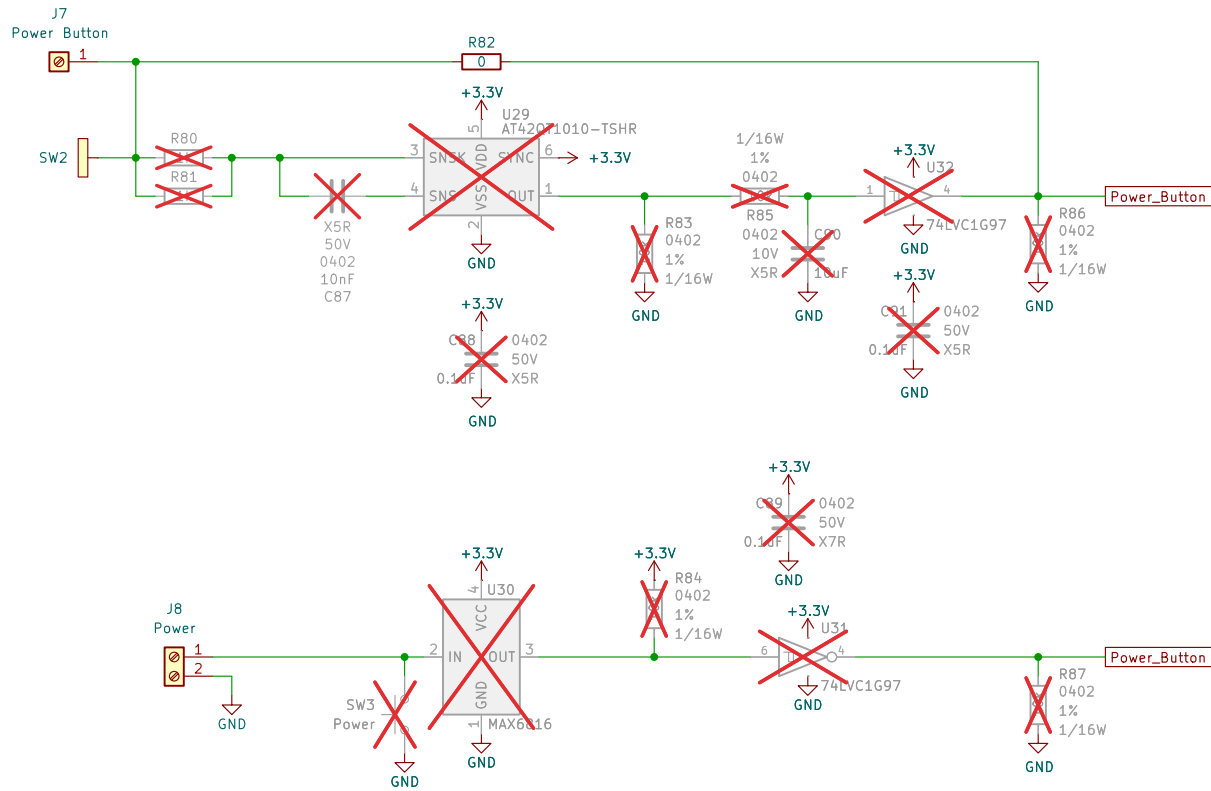
Sheet: /LED Panel Connectors/
File: led_panel_connectors.kicad_sch

Title:

Size: ADate:KiCad E.D.A. 10.0.3

Rev:Id: 18/20

Sheet: /LED Panel Connectors/ File: led_panel_connectors.kicad_sch		
Title:		
Size: A	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 18/20



Sheet: /Pushbuttons/
File: pushbuttons.kicad_sch

Title:

Size: A
KiCad E.D.A. 10.0.3

Date:

Rev:

Id: 19/20

